MIL-C-36417(DM)
22 March 1965

MILITARY SPECIFICATION

CELLS, SALINITY INDICATOR

1. SCOPE

1.1 Scope. This specification covers indicator cells, used to determine the salinity of water.

1.2 Classification. Salinity indicator cells shall be of the following types, as specified (see 6.1):

Type I. - CN14 - RPL.

Type II. - CN9 - N - RPL.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-N-836	Nut, Plain: (Hexagon, Square, Cap and Welding) Nut, Slotted, and Castellated, Hexagon.
FF-S-85	Screw, Cap, Slotted - and Hexagonal Head.
FF-S-86	Screw, Cap, Socket-Head.
FF-S-92	Screws, Machine: Slotted, Cross-Recessed or Hexagonal Head.
QQ-B-613	Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet, and Strip).
QQ-B-626	Brass, Leaded and Non-Leaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).

FSC 6630

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PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-R-880	Rubber, Medium-Soft (for Airport, Airtight and Watertight Door, and Hatch Gaskets).
MIL-C-915	Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-F-1183	Fittings, Tube, Cast Bronze, Silver-Brazing.
MIL-V-1189	Valve, Gate, Bronze, Flanged Ends, Working Pressures up to 400 P.S.I. (Shipboard Use).
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-P-15024	Plates, Identification -- Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-S-15103	Salinity Indicating Equipment.
MIL-M-15176	Mica (Extending Pigment).
MIL-S-16410	Steel, Spring, Bars and Wire.
MIL-W-16878/6A	Wire, Electrical, Type ET, 200 C. and 260 C., 250 Volts, (Insulated, High Temperature) (Navy).

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.

DRAWING

Navy Dept. - Bureau of Ships.

S6505/H/3158366D	Salinity Indicating Equipment Type 1C/CN14 Cell and Valve Assembly.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

3.2 Construction. The salinity indicator cell shall consist essentially of a cell tube fitted with a packing nut, cable gland nut, and electrode assembly. A temperature compensator and a flexible three-wire cable shall be connected to the electrode assembly. The type I and type II salinity indicator cells shall be the 1C/CN14 style, constructed in strict accordance with Drawing 86505/H/3158366, Revision D3, and shall consist of the following piece numbers:

Type I	* Type II
(a) 1 to 10 inclusive	(a) 1 to 21 inclusive
(b) 12 to 21 inclusive	(b) 21A
(c) 21A	(c) 22 to 25 inclusive
(d) 22 to 25 inclusive	(d) 28 - 29 inclusive
(e) 28	(e) 34 - 38 inclusive
(f) 34 - 38 inclusive	(f) 40 - 45 inclusive
(g) 40 - 45 inclusive	

* Piece No. 11 shall be Manufacturer's Piece No. PM-4N instead of PM-3N. Superseded specifications listed in the drawing and the applicable replacement specifications for the required pieces numbers are as follows:

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(a) MIL-B-895 superseded by QQ-B-613 and QQ-B-626.
(b) MIL-P-14 superseded by MIL-M-14.
(c) MIL-S-933 superseded by FF-S-85, FF-S-86, FF-S-92, and FF-N-836.
(d) MIL-I-15024 superseded by MIL-P-15024.
(e) MIL-46-S-53 (INT) shall not apply.

3.2.1 Electrode assembly. The elctrode assembly shall consist of an inner and an outer electrode constructed of brass. The electrode shall be concentric and the spacing shall be positive. The exterior surface of the inner electrode and the interior surface of the outer electrode shall be sheathed with nonporous platinum. The platinum sheathing shall be 0.003 inch ± 0.0005 inch thick, and the total active area of the platinum sheathing on the electrodes shall not be less than 3 square inches. The platinum sheathing shall be silver soldered to the outer electrode and soft soldered to the inner electrode. The end of the outer electrode that meets the cell tube adapter shall be provided with a knurled brass collar which is an integral part of the electrode. The knurled collar shall be 1-5/32 inches in outside diameter and 1/8 inch wide, nominal dimensions, and shall make a flush fit with the cell tube adapter. The outer electrode shall contain four holes, each of 1/4-inch nominal inside diameter. The holes shall be machined through the cylindrical side walls of the electrode and shall be equally spaced around the circumference of the electrode. The center of each hole shall be approximately 5/8 inch from the free end of the outer electrode. When the salinity indicator cell is in operation, the current flowing between the electrodes shall not come in contact with or be exposed to any metal other than platinum. The electrodes shall be in direct and maximum contact with the full and unimpeded flow of the water stream when they are fully inserted in the working position.

3.2.2 Temperature compensator. A temperature-compensating element shall be attached to the inner electrode and shall provide automatic temperature compensation for variations in the temperature of the water surrounding the electrodes.

3.2.3 Packing nut. The packing nut shall contain a small seep hole of approximately 1/16-inch inside diameter, drilled through its radial depth. The seep hole shall give a warning seepage of fluid if an attempt is made to remove the indicator cell from its working position before closing the water supply valve. The packing nut shall contain a setscrew which screws down into a groove in the cell tube to prevent axial displacement of the cell tube by hydrostatic pressure.

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3.2.4 Cell tube. The cell tube shall be fabricated of seamless brass tubing (QQ-B-613 and QQ-B-626 shall not apply). The cell tube shall be fitted with the electrode assembly on the end which is to be inserted into water. The opposite end of the cell tube shall contain the cable gland nut which clamps the three-wire cable in the cell tube. The groove which controls the insertion distance of the electrode shall be positioned as required for the specific type.

3.2.5 Retaining ring. The retaining ring shall be fabricated of berrylium-copper (MIL-46-S-53 (INT) shall not apply).

3.2.6 Insulation. All insulation and plastic parts, with the exception of the flexible cable, shall not be damaged or affected by 500° Fahrenheit (F.) heat.

3.2.7 Positioning. The salinity indicator cell shall be readily inserted and removed from its operating position safely and positively.

3.3 Range. The salinity indicator cell shall have a range of 0.0 to 1.5 equivalent parts per million of chloride for temperatures from 40° F. through 250° F.

3.4 Cell rating. The salinity indicator cell shall have a cell constant of 0.1 and shall operate up to 125 pounds per square inch hydrostatic pressure over a temperature range of 40° F. to 250° F.

3.5 Performance.

3.5.1 Calibration. The salinity indicator cell shall be calibrated to ± 1.0 percent of the full scale reading, when tested as specified in 4.4.1.

3.5.2 Accuracy. The salinity indicator shall be accurate to ± 3.0 percen of the full scale reading, when tested as specified in 4.4.2.

3.5.3 Response time. The response time of the temperature compensator shall not be greater than 15 seconds, when tested as specified in 4.4.3.

3.5.4 Endurance. The resistance between the electrodes shall be 50 megohms or greater, when tested as specified in 4.4.4.

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3.5.5 Dielectric strength. The salinity indicator cell shall successfully withstand without breakdown of any parts, the application of 900 volts, 60 cycles, for 1 minute, when tested as specified in 4.4.5.

3.6 Identification marking. The salinity indicator calls shall be marked in accordance with the contractor's commercial practice and shall include, as a minimum, the following:

(a) The Federal stock number.
(b) Item identification.
(c) Manufacturer's name and part number.

Type I cells shall be marked "Replacement Part for Cell Type CN14" or similar statement. Type II cells shall be marked "Replacement Part for Cell Type CN9-N" or similar statement. The identification marking shall be placed on the packing nut, piece No. 13, of the drawing. The method of identification marking specified in the drawing shall not apply.

3.7 Workmanship. Workmanship shall be first class throughout. The salinity indicator cells shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

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	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the cell.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Calibration	3.5.1	4.4.1	100%	*
Accuracy	3.5.2	4.4.2	S-3	*
Response time	3.5.3	4.4.3	S-3	*
Endurance	3.5.4	4.4.4	S-3	*
Dielectric strength	3.5.5	4.4.5	S-3	*

* Acceptance number shall be zero.

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4.3 Examination. The indicator cells shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major	
101	Component part missing.
102	Outer electrode not properly sheathed with platinum.
103	Inner electrode not properly sheathed with platinum.
104	Cable wires not properly labelled.
105	Cable wires incorrectly colored.
106	Outer electrode not furnished with four radial holes.
107	Component parts do not fit properly.
108	Packing nut not furnished with seep hole.
109	Cell tube indicator groove missing or not filled with red paint.
Minor	
201	Cell not free of grease, dirt, and foreign matter.
202	Marking not present, correct, legible, or permanent.

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The indicator cells shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Calibration. The salinity indicator cell shall be calibrated against a standard salinity indicator cell which is traceable back to the National Bureau of Standards. The calibration shall be carried out in 122° F. water having 0.15 equivalent parts per million of chloride. The indicating meter shall have a scale which is not less than 3.2 inches long (measured on the arc) and the metering circuit shall be the same as under normal operating conditions.

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4.4.2 Accuracy. Testing shall be as specified in 4.4.1 except that the tests shall be run at 50° F., 150° F., and 250° F., using water having 0.02, 0.15, and 1.5 equivalent parts per million chloride at each temperature.

4.4.3 Response time. Two temperatures which are at least 122° F. apart shall be selected as the extreme temperatures for the test. The higher temperature shall be designated as TH and the lower as T_L. The mean temperature shall be designated as T_M and shall be calculated as follows:

T_M = 0.63 percent T_H + 0.37 percent T_L.

The resistance of the thermally responsive element within the indicator cell shall be determined at T_M either by measurement or by calculation using table V of MIL-S-15103 for 0.231 EPM. The thermally responsive element within the cell shall then be connected to a suitable resistance circuit and the completely assembled indicator cell, with the electrodes open circuited, shall be allowed to stabilize in water maintained at T_L. The cell shall then be rapidly immersed in water maintained at T_H. Using a stopwatch, the elapsed time, from immersion of the cell in the water at T_H until the resistance corresponding to T_M is reached, shall be determined. The elapsed time shall be the response time of the indicator cell.

4.4.4 Endurance. The salinity indicator cell shall be exposed to the following conditions applied consecutively and continuously without interruption:

(a) Hydrostatic pressure of 600 pounds per square inch at a temperature of 60° F. to 80° F. for 5 minutes.

(b) Hydrostatic pressure of 250 pounds per square inch at a temperature of 285° F. for 1 hour.

(c) Water vapor at 212° F. for 1 hour.

(d) Hydrostatic pressure of 250 pounds per square inch at a temperature of 60° F. to 80° F. for 1 hour.

The resistance between the electrodes shall be determined immediately upon completion of the final exposure period.

4.4.5 Dielectric strength. The test shall be as specified in method 301 of MIL-STD-202. The voltage shall be applied between the insulated portions and ground.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) cell, of the type specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging.

5.1.2.1.1 Electrode assembly. The electrode assembly shall be completely wrapped in suitable waterproof paper, overwrapped in corrugated fiberboard or other cushioning material as required for necessary protection against shifting or damage, and packaged in a carton of appropriate size.

5.1.2.1.2 Other components. The gland nut, flange, and cell tube shall be wrapped in corrugated fiberboard or other suitable cushioning material to prevent shifting or damage, and further packaged in a carton of appropriate size.

5.1.2.1.3 Unit package. Each unit, packaged as specified in 5.1.2.1.1 and 5.1.2.1.2, shall be overpackaged in a carton of appropriate size constructed in accordance with PPP-B-636. Carton closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Units, packaged as specified in 5.1.2.1.3, shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Gross weight of the exterior container shall not exceed the weight limitation of the applicable box specification, and in no case shall exceed 200 pounds.

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5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with the appendix of the applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of PPP-B-636. In addition, waterproofing shall be as specified in 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Units, packaged as specified in 5.1.2.1.3, shall be overpacked in an exterior container conforming to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Gross weight of the exterior container shall not exceed the weight limitation of the applicable box specification, and in no case shall exceed 200 pounds.

5.2.3 Level C. The commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Preparing activity:
Defense Supply Agency. - DM

Review activities:
Army. - MD
Navy. - MS
Air Force. - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION MILITARY SPECIFICATION CELLS, SALINITY INDICATOR MIL-C-36417(DM) dated, 22 March 1965

ORGANIZATION (Of submitter)

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)

DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

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DEFENSE MEDICAL SUPPLY CENTER
ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE DMS)
3RD AVENUE AND 29TH STREET
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